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The East Antarctic Sea Ice Zone: Ice Characteristics and Drift

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ABSTRACT: Results from studies of the surface energy balance and the ocean structure in the presence of fast ice near Mawson on the Antarctic coast are used to illustrate the important ways in which sea ice interacts with the ocean and atmosphere. Away from the coast, ship and drifting buoy observations are used to characterize the E Antarctic sea ice zone in a study area between 60° E and 120° E and S of 61° S. Divergent drift over most of the region plays a dominant role in expanding the ice extent in autumn and in determining the characteristics of the pack. Much of the sea ice in the region is young thin ice which forms in leads and polynyas, and in late spring in the study area, the ice thickness averaged over the total ocean surface within the ice edge less than 0.4 m. Even in winter the majority of ice floes off E Antarctica are probably less than 1 m thick.

Introduction

Sea ice has a marked influence on the interaction between high latitude ocean and atmosphere and hence anomalies and variations in sea ice extent have long been considered potential modifiers of both regional and global climate. An ice cover markedly changes the surface albedo and roughness, insulates the ocean from the atmosphere, and redistributes salt in the ocean by brine rejection during ice growth. Hence it affects the radiant and turbulent exchanges of energy, the transfer of momentum, and the vertical mixing and salt distribution in the ocean. The sea ice extent varies not only just over a distinct annual cycle but also shows noticeable interannual changes.

While satellite data have greatly increased our knowledge of Antarctic sea ice, a full understanding of the climatic role of ice also requires knowledge of detailed ice processes and characteristics that cannot be determined from satellites. Considerable research into sea ice processes and characteristics has been conducted in the Arctic; however the interior of the Antarctic seasonal sea ice zone is an area where the ice characteristics and their variability (which are both temporally and spatially large) are little known. The contrasting geography of the two polar regions leads to major thermodynamic and dynamic differences in their sea ice zones. In the Antarctic, the sea ice is mostly first year ice with a thickness typically less than 2 m, and the area covered by the ice has a large annual variation (4×10^6 to 20×10^6 km^2). At the outer boundary the ice is unconstrained by land, divergent and subject to meridional advection, and free to interact with the open ocean. In the Arctic Ocean much of the ice is multi-year with a mean thickness of about 4 m. There is a marked longitudinal asymmetry in the land-sea distribution, and the seasonal ice range is much less than in the Antarctic Ocean (7×10^6 to 14×10^6 km^2) due to the land-locked nature of the Arctic Ocean. The marginal ice zone outside the central Arctic basin is not subject to the unlimited fetch of winds and currents that occurs around the Antarctic and there are also large differences in the structure and stratification of the Arctic and Southern Oceans. Hence climatic inferences for S sea ice cannot be directly based on measurements and observations made in the Arctic.

In this paper sea ice interactions with atmosphere and ocean are illustrated by a summary of results from process studies over land-fast ice near Mawson, Antarctica. New data are then presented on the drift and characteristics (particularly ice type and ice thickness) of the pack ice in the Indian Ocean sector of the Southern Ocean.

Ice Extent, Ice Concentration and Ice-Weather Relationships in the Antarctic

Although ship observations have long been used in compilations of sea ice distribution, it is only since the advent of remote sensing from earth orbiting satellites that an accurate assessment of Antarctic sea ice distribution, and its variation, has been obtained. Initially, from the mid-1960s until the early 1970s, most ice data were obtained with visible and infrared satellite imagery with relatively poor spatial resolution. Gross ice boundaries during the long polar night were obtained from thermal infrared imagery with cloud filtering achieved by comparing images for several days and using composite minimum brightness techniques (McLain 1978)

The development of passive microwave sensors, such as the Electically Scanning Microwave Radiometer (ESMR) deployed on Nimbus 5 in 1972, greatly improved ice mapping capability irrespective of cloud or darkness. Sea ice extent at a resolution of better than 40 km can be monitored year round with microwave radiometers (Zwally and Gloersen 1979). It is also possible to derive the important parameter of ice concentration to about ± 15% from the microwave brightness temperatures (Comiso and Zwally 1982). Improved ice concentration information and additional data on sea ice surface temperature, ice type, and the percentage of thin ice would be obtainable from the multispectral microwave radiometers on Nimbus 7 and DMSP satellites provided that the variations of the microwave signature of sea ice with age, thickness, salinity, structure, and surface characteristics were understood more fully than at present (e.g. Comiso and others 1984).

Zwally and others (1983) estimated the four year mean ice extent for 1973 to 1976 derived from Nimbus 5 ESMR data. The total ice extent varied from a minimum of approximately 4×10^6 sqkm in February to a maximum of 20×10^6 sqkm in September. For the Pacific and Indian Ocean sectors (20° E to 160° E) the ice extent varied seasonally from 1 to 6.3×10^6 km^2. Their analysis of the areas of ice of different concentration showed that, at maximum ice extent, only about 75% of the ocean surface within the seasonal sea ice zone is actually covered with ice.

This open nature of the Antarctic pack greatly modifies the air-sea interaction. Early estimates of the annual heat loss over the S sea ice (e.g. Fletcher 1969) assumed a continous ice cover but Weller (1980), using satellite data of ice extent and concentration, concluded that the energy losses of the Southern Ocean might be greater than Fletcher's estimates by up to a factor of six, and that the energy balance in the seasonal sea ice zone remains the major uncertainty in our understanding of atmospheric thermal forcing in the Southern Hemisphere. While recent ship-based measurements (Andreas and Makshtas 1985) have provided some data on turbulent and radiative fluxes over the interior pack ice of the Southern Ocean, the Antarctic pack ice zone remains logistically difficult to study in detail compared to the shore-fast ice. Data on the seasonal variation of the energy balance of fast ice can, however, provide information on the relative importance of various energy-exchange processes for various categories of ice thickness and type and are discussed in the next section.

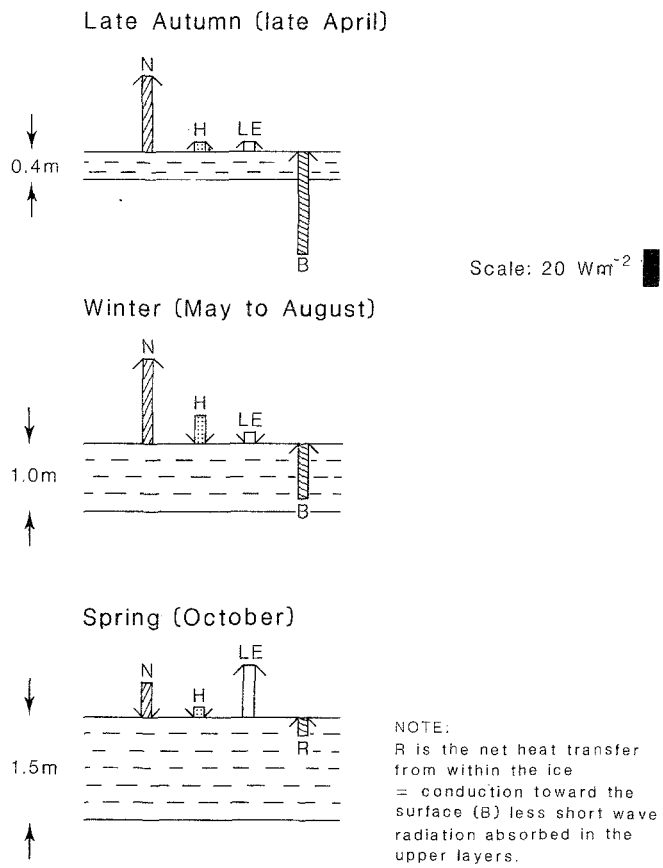
Data sets for monitoring the seasonal cycle and interannual variations of Antarctic ice extent have been compiled either directly from the satellite data, or from synoptic ice charts, such as those prepared by the US Navy/NOAA Joint Ice Centre (Godin 1979), which utilize a variety of satellite and other data when available. The sea ice data sets range from those recording only ice extent (e.g. Jacka 1983; Lemke et al. 1980) to those also recording ice concentration and distribution (Zwally et al. 1983; Naval Oceanography Command 1985). Lemke et al. (1980) showed that interannual variations of ice extent are generally of the order of 1/3 of the seasonal variations and that the longitudinal zones of high seasonal variability (for example the Ross and Weddell Seas) are also the zones of high interannual variability. Sturman and Anderson (1985) compared seven such Antarctic sea ice data sets, and concluded that they indicate similar long term trends in ice extent since 1973, but that different analysis techniques give varying estimates of the mean and the annual cycle of the ice covered area.

Such data sets have been used in a number of studies to investigate the relationship between sea ice extent and concentration, and either individual synoptic weather events or mean conditions. For example Schwerdtfeger and Kachelhoffer (1973) reported a statistical correlation between the ice margin and the latitudinal band of maximum cyclone frequency during autumn and spring, and Streten and Pike (1980) showed that the 5 year means of zonal westerly wind speeds to the N of the Antarctic increased with ice extent, although there was no clear correlation for individual years. Carleton (1981) suggested that greater ice extent tended to increase regional cyclogenesis in the ice edge zone and noted that cyclonic activity may enhance ice advection. Ackley and Keliher (1976) examined the positive feedback between the atmosphere and sea ice in the Bellingshausen Sea in the 1973 winter when a rapid ice decrease possibly caused a gain in atmospheric heat content, thereby augmenting cyclone activity. Similarly Parkinson and Cavalieri (1982) noted strong indications that the large scale features of interannual ice differences, such as the tendency for large regions of positive ice anomaly to be balanced by large negative anomalies elsewhere, often resulted from interannual changes in the positioning and internsity of cyclonic and anticyclonic systems. Streten (1983) analyzed zonally averaged and regional Antarctic sea ice extent during the FGGE year and concluded that the track, intensity and frequency of cyclones appeared to be the most important factor affecting the development and maintenance of anomalies in ice extent; Budd

(1986) found a similar result from a 10 year analysis of S Hemisphere cyclones. Cavalieri and Parkinson (1981) used Fourier decomposition of the ice edge, pressure and temperature fields to define the dominant spatial scales for each variable during the course of the year. Their results demonstrated that ice-atmosphere coupling varied in strength throughout the year and while the ice edge responded to large-scale pressure and temperature variations, there was also evidence for an atmospheric response to a changing ice cover.

Lemke et al. (1980) studied the pattern of sea ice variability using empirical orthogonal functions (EOFs). They found that in the Antarctic the longitudinal EOFs of ice extent showed that most of the variance was contained in patterns representing anomalies of opposite sign at different longitudes. The main contribution to the first EOF came from the Weddell Sea, whereas the second EOF was mainly determined by the Ross Sea, suggesting that these two principal ice production areas are largely decoupled. In East Antarctica the greatest variability occurred in the Prydz Bay region (70° E to 80° E).

While diagnostic studies provide evidence of the links between sea ice and weather, shortcomings in data of Antarctic ice conditions and processes constrain the confidence that can be placed on the results of these analytical studies. Such studies use large-scale features to infer small scale effects. Further insight into these effects can be gained from process studies such as those exemplified in the following.



N = net radiation flux; H = turbulent sensible heat, LE = turbulent latent heat, B = conducted heat

Fig 1 Schematic representation of the seasonal changes in surface energy fluxes over land-fast sea ice of different thickness.

Surface Energy Balance and Ocean Structure in the Fast Ice Zone

A study of ocean-ice-atmosphere energy exchange and its seasonal changes in the coastal fast ice zone was undertaken near Mawson (67° 36' S, 62° 53' E) over a number of years. In this study energy fluxes over open water or ice were measured by standard micrometeorological profile techniques using instruments mounted on either moored rafts or towers. Datails of the program and the results have been reported by Weller (1968), Allison and Akerman (1980), and Allison et al. (1982), and the seasonal changes are summarized below.

In mid February, more than one month before sea ice formation at this site, the ocean loses around 10 MJ m^{-2} day^{-1} mostly from turbulent transfer of sensible and latent heat. Both net radiative and turbulent loss increase in March until an ice cover is formed, changing both the surface albedo and acting as a physical barrier between the warm ocean and cold atmosphere. Allison and Akerman (1980) measured a decrease in the total turbulent energy loss from around 120 Wm^{-2} over open water to less than 10 Wm^{-2} over 0.25 m thick ice in March. The net radiation however initially only changes slightly, partly because incoming short wave radiation is offset by a decrease in outgoing longwave radiation since the ice surface is more than 10° C colder than the sea surface. Heat loss from thin ice remains high as illustrated by the model results of Maykut (1978) which showed that nearly twice as much heat is lost due to turbulence in autumn from ice 0.05 m thick than from 0.40 m thick ice and that in winter the difference is even greater.

The changes in surface energy balance as the ice continues to thicken between late April and October are shown schematically in Fig 1. In late April, when the ice cover is less than 500 mm thick, the ice loses about 50 Wm^{-2} by net radiation and about 10 Wm^{-2} by turbulent transfer. This energy loss is balanced by conduction through the relatively thin ice. Once the ice grows beyond about 500 mm thick the conducted heat flux decreases appreciably, a surface inversion develops above the ice and the sensible heat flux is from atmosphere to ice. The latent heat transfer may also be towards the surface, resulting in hoar frost deposition. The winter net radiative loss remains fairly constant at 40 to 50 Wm^{-2} and is balanced by the energy gain from turbulent transfer and conduction through the ice. Although the average total turbulent energy gain is

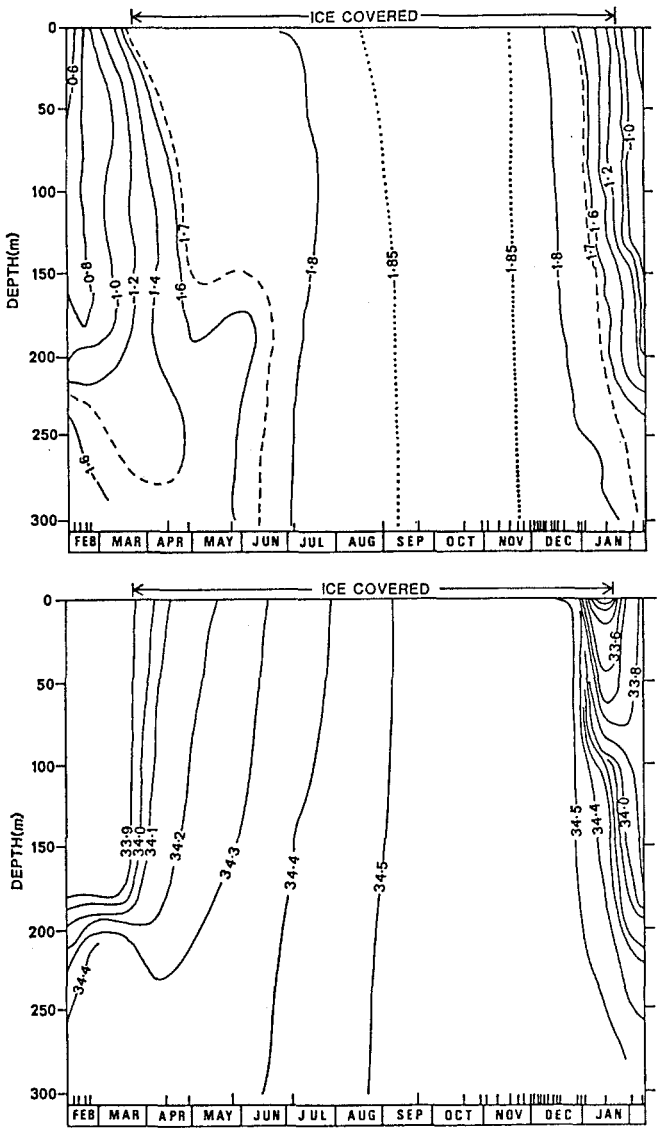


Fig 2 Isopleths of ocean temperature (° C, top) and salinity (parts per thousand, bottom) under first year fast ice

generally less than 20 Wm^{-2}, periods of high turbulent energy gain may occur during blizzards. Andreas and Makshatas (1985) also noted the importance of the large-scale wind strength and direction in determining total turbulent heat loss over Antarctic sea ice.

By early October the net radiation loss decreases and by mid October becomes a net radiative gain (about 15 to 25 Wm^{-2}) as the incoming short wave radiation increases. Heat loss occurs through ablation and the turbulent latent heat transfer is now away from the surface (up to 30 Wm^{-2}) although a small sensible heat transfer (about 5 Wm^{-2}) from the atmosphere to the ice may continne. Shortware radiation absorption within the upper layers ot the ice starts to become important if there is no snow cover, and the conducted heat flux towards the upper surface increases, although the net energy supplied to the surface from within the ice (conduction less absorbed shortwave radiation) is only 5 to 10 Wm^{-2}. Between early and mid-November, unstable conditions develop over the ice and the sensible heat flux is again directed away from the surface.

Brine rejected by a growing ice cover and energy exchange at the lower ice surface influence the structure of the upper ocean as shown in Fig 2. These data show the measured seasonal change in ocean temperature and salinity at the Mawson site and are discussed in more detail by Allison et al. (1985). The upper ocean loses heat rapidly in late February but remains stratified until after the formation of the ice cover in mid March to early April. The pycnocline is then rapidly broken down by haline convection as brine is rejected from the ice. The salinity of the total column steadily increases while the temperature decreases as the ice cover grows to its maximum thickness of 1.8 m, and by September the water mass is isothermal with a constant salinity.

Initially during the period of ice decay (early November to mid December) there is little change in the salinity of the water column under the ice although the temperature increases slowly during this time. From the second half of December through to the end of January there is a large increase in the temperature of the column and a decrease in the salinity.

The decrease in salinity is in part due to ablation at the bottom of the ice sheet, but is due mainly to freshwater runoff from the Antarctic ice sheet close to the coastal site. After the ice breaks out in late January there is little change in salinity but the freshwater input from the ice sheet is well mixed by strong winds (typically during events with gusts to 40 ms^{-1}) and a summer pycnocline forms again at about 200 m depth.

Consideration of the salt and heat budget of the water column shows that a net advection of salt, but little advection of heat, to the site occurs during the winter months. From November onwards there is an increasing oceanic advection of heat to the region, although the salt budget at this time is dominated by freshwater input from the Antarctic ice sheet. The oceanic heat flux from the water to the ice increases at the same time as seen also from ice thickness changes at several other coastal Antarctic sites (Allison 1981). The increase in available heat is possibly related to the retreat of the Antarctic sea ice further N and the late spring reduction of ice concentration.

Ice Type and Thickness in the Late Spring Pack

The results in the previous section, although obtained for fast ice, illustrate the importance of ice thickness and open water extent in determining Antarctic ocean-atmosphere heat exchange and, via the ice growth rate, the salt flux into the ocean. Yet ice thickness cannot be measured directly from satellites and there are few available data on ice characteristics within the Antarctic sea

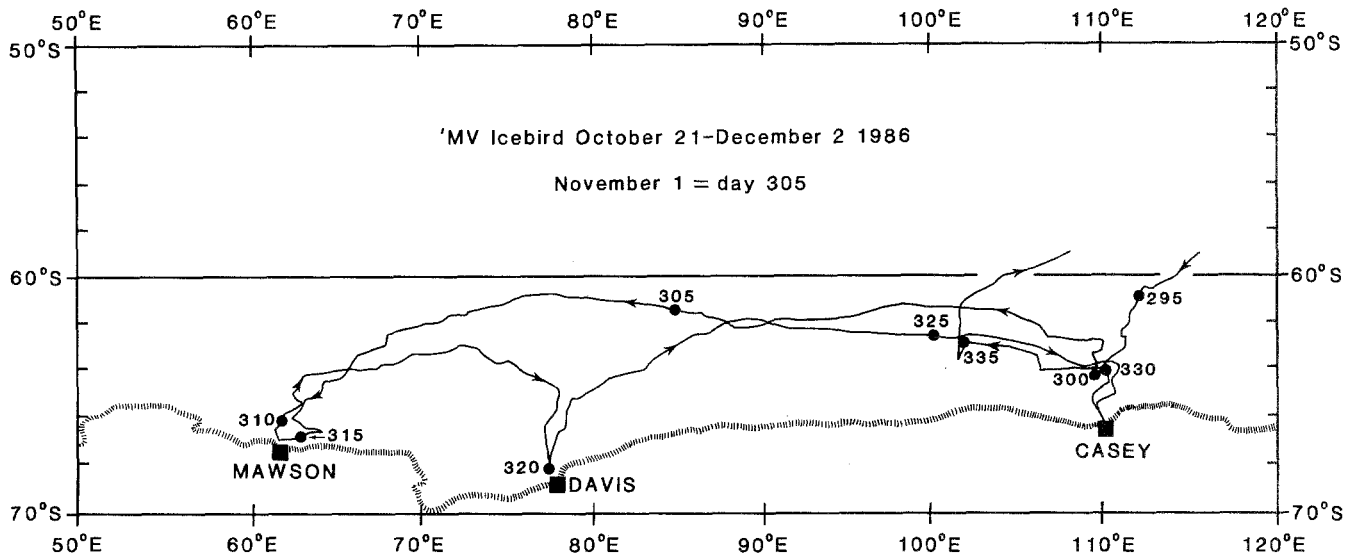


Fig 3 Cruise track of MV 'Icebird', 1986

ice zone. With generally divergent currents over most of the region, we expect to find considerable open water or thin ice.

Ackley et al. (1983) reported on physical properties of the late spring pack in the Weddell Sea, and from ice floes sampled during that program Gow et al. (1982) found that much of the ice consisted of aggregated frazil crystals which they suggested formed as a result of brine convection induced by rapid freezing of sea water in leads and polynyas. For the E Antarctic sector between 40° E and 80° E Jacka et al. (1987) observed ice conditions in the interior pack shortly after the time of maximum ice extent, and before significant summer decay commenced. They also found a large fraction of frazil in the ice column, a mix of floes of different ages with considerable thin ice, and a relatively uniform ice field with rafting mostly confined to ice less than 0.5 m thick. By December most thin ice had melted and the pack had become much more homogeneous.

The observations presented here were made in 1987 at a similar time of the year (late October and November) to those of Jacka et al. (1987) but cover a wider area of the E Antarctic pack. Visual observation of the concentration of different ice types, floe size, ridging, and estimates of predominant ice thicknesses and snow cover were made every three hours from the bridge of MV 'Icebird'. For thin ice types the thickness was estimated from the grayness of the ice while for first year and snow covered ice it was visually estimated to an accuracy of about 20% from floes overturned by the ship's passage. Although the individual observations are not quantitative a statistical compilation of these data provides a general picture of ice characteristics in late spring. The cruise track of 'Icebird' (basically two zonal traverses along 60 to 64° S with N-S transects into the coast around 62° E, 70° E and 110° E), is shown in Fig 3. It is stressed that the cruise was not primarily for scientific purposes but a re-supply voyage to several Antarctic stations. The route, and hence the observations, may therefore be somewhat biased towards lower ice concentrations by the requirement to find the easiest passage and by the vessel's limited ice breaking capability. No estimate can be made of the magnitude of any such bias, but route selection was generally determined using observations made from the vessel and short range helicopter flights. No larger scale differences between ice concentrations along the route and in neighbouring regions were identifiable from low resolution NOAA APT imagery. Observations during the helicopter flights were also used to check the local uniformity of ice conditions and to correct for any bias due to viewing angle from the ship's bridge.

Fig 4 and 5 show the latitudinal variation of the floe size, visually estimated thickness of the predominant ice type, and total ice concentration along the essentially N-S parts of the route. There were regional differences in ice characteristics but more importantly, significant changes with time. On the W side of Prydz Bay (62–78° E) (Fig 4a) total ice concentration increased from 2–6/10 in the first 250 km to a mean of around 7–8/10 in the interior pack. Floe sizes increased steadily as the concentration increased but the thickness of the presominant first year floes was fairly constant around 0.9 m. The Prydz Bay region experiences a clockwise gyre circulation (Smith et al. 1984) and the E side (Fig 4b) had more variable ice thickness and generally smaller floe sizes. Again the concentration increased from 3/10 to about 8/10 moving southward but S of 67° S there was an extensive shore polynya covered with new, thin ice.

The section along about 110° E was observed both in late October and late November. In late October

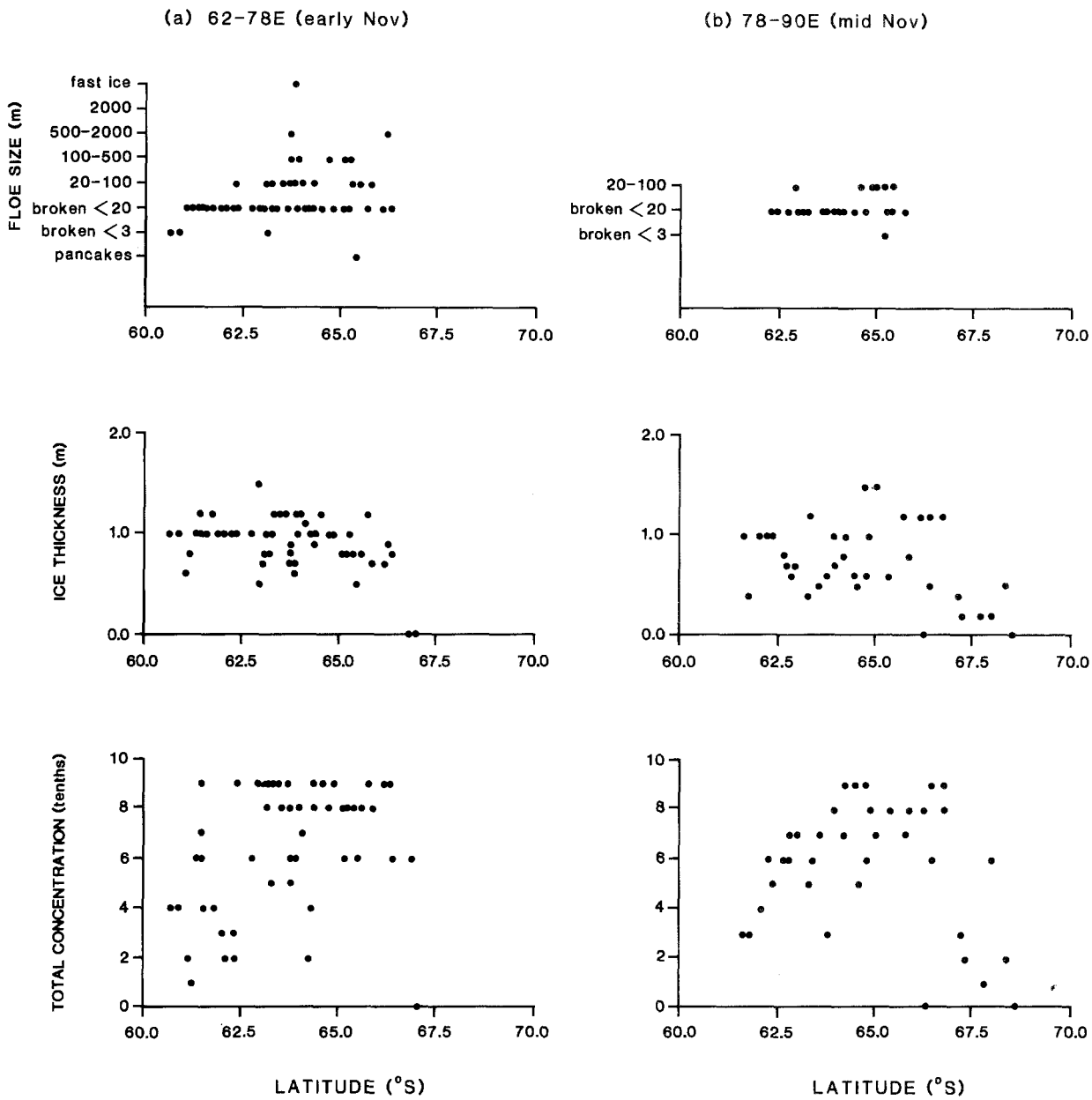


Fig 4 Latitudinal variation of floe size, ice thickness and total ice concentration for 62–78° E (a) and 78–90° E (b)

(Fig 5a) the ice concentration, thickness, and floe size all increased steadily from the ice edge to 64° S, beyond which point the ship was unable to penetrate because of 10/10 concentration of largely unbroken first year ice. One month later however (Fig 5b), there were significant changes. The ice edge had retreated about 200 km and while ice thickness and concentration still increased with latitude they had decreased by about 0.3 m and 2/10 respectively from the October values at the same latitudes. The more open pack also resulted in floes breaking up and the floes were a lot smaller and more uniform than in October.

The observations incluced the percentage concentration of different ice types, each representing a different thickness category. Ice types characterized were new ice (< 0.05 m) nilas (< 0.1 m), young grey ice (0.1–0.15 m), grey white ice (0.15–0.30 m), first year ice less than 0.7 m, first year ice less than 1.2 m and first year ice greater than 1.2 m thick. From these observations the area-averaged ice thickness has been calculated as the sum of a representative thickness for each category, weighted by the proportional concentration for each category. The latitudinal variation of this weighted thickness is shown in Fig 6 for all observations, both for

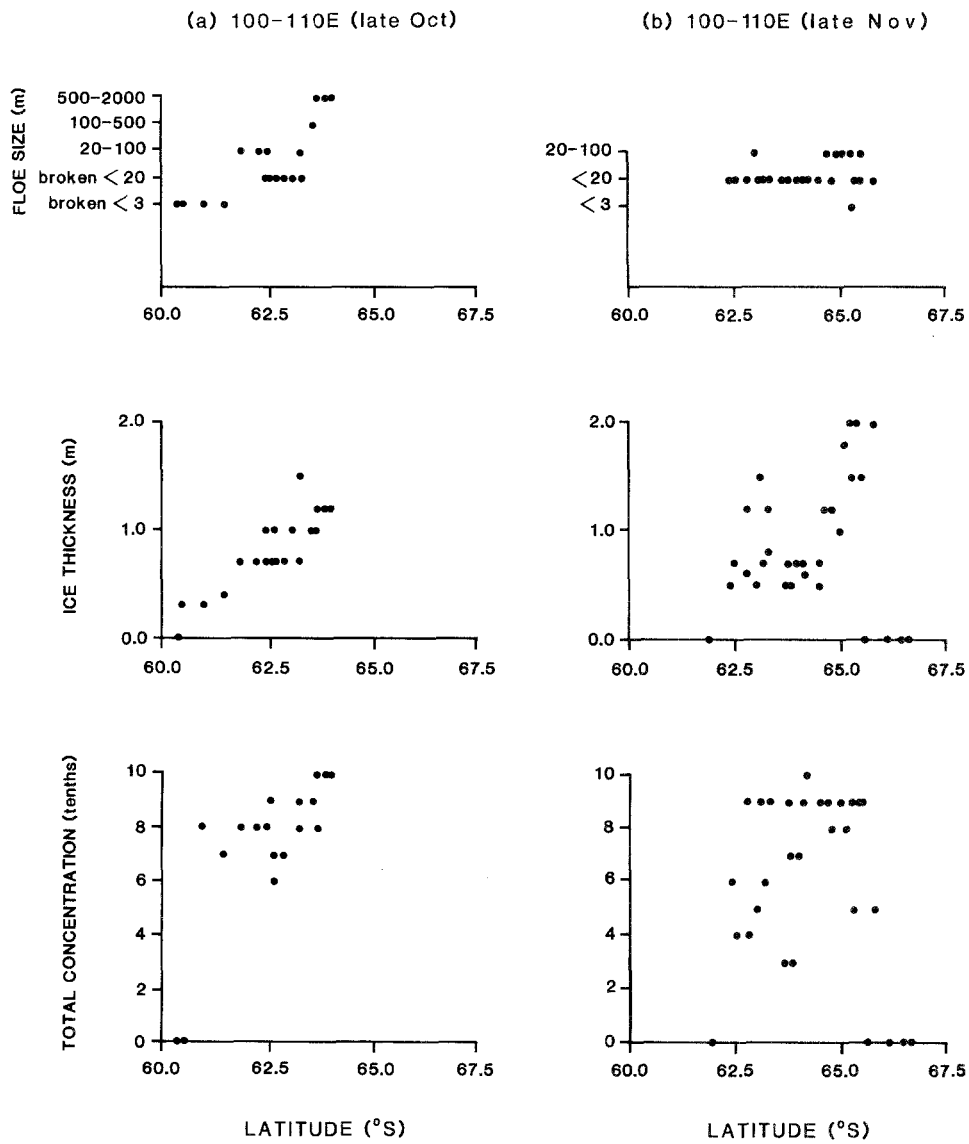
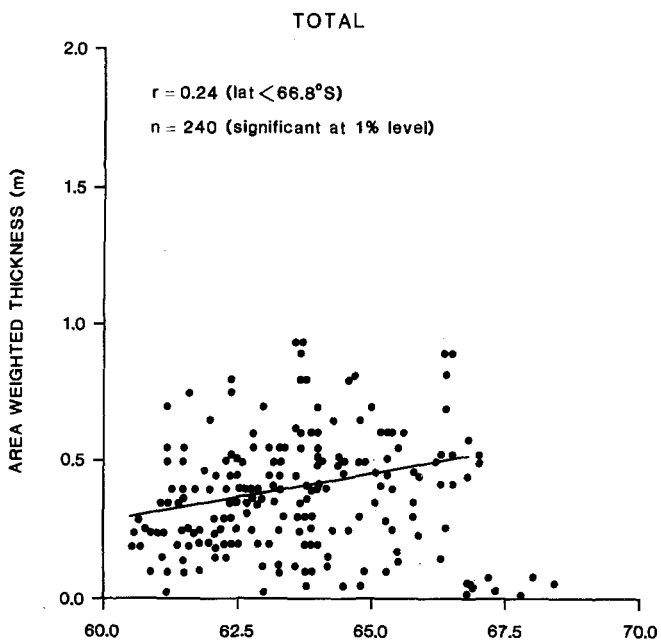


Fig 5
Latitudinal variation of floe size, ice thickness and total ice concentration for 100–110° E in late October (a) and in late November (b)

the overall ice thickness (weighted for open water) and for the weighted thickness of the ice covered area only. For latitudes N of the shore polynya zone there is a wide scatter, but a statistically significant increase of total area weighted thickness with latitude. The weighted thickness of the ice covered area only does not show a similar statistically significant increase, indicating that the total ice mass increases with latitude because of increasing ice concentration rather than the presence of thicker ice categories. For the period of observation (October 21 to December 2) the weighted total ice thichness was 0.37 m and the average thickness of ice over the ice covered area only was 0.75 m.

Fig 7 shows the area weighted thickness of new ice only. Although ice concentration was generally greater and there was less open water at higher latitude, there was still more new ice formation there because of lower temperatures. Fig 8 shows the air and sea surface temperatures during the cruise and the periods when new ice and other this ice categories were observed. New ice was only observed to form when the air temperature was below about −4° C, no new ice was seen after late November, and the ice became more homogeneous in late November and early December as thin ice types melted.

The observations have been smoothed zonally to derive averages, of the weighted ice thickness and total ice concentration every 5 day and 1° of latitude. These averages have been used to produce the isopleths in Fig 9, showing broad changes with latitude and time. The ice edge shown in Fig 9 is the mean ice edge between 50° E and 120° E derived from the US Navy/NOAA Joint Ice Center (JIC) charts which at this time of year are based predominantly on visible and microwave satellite data. The top sections of Fig 9 show total



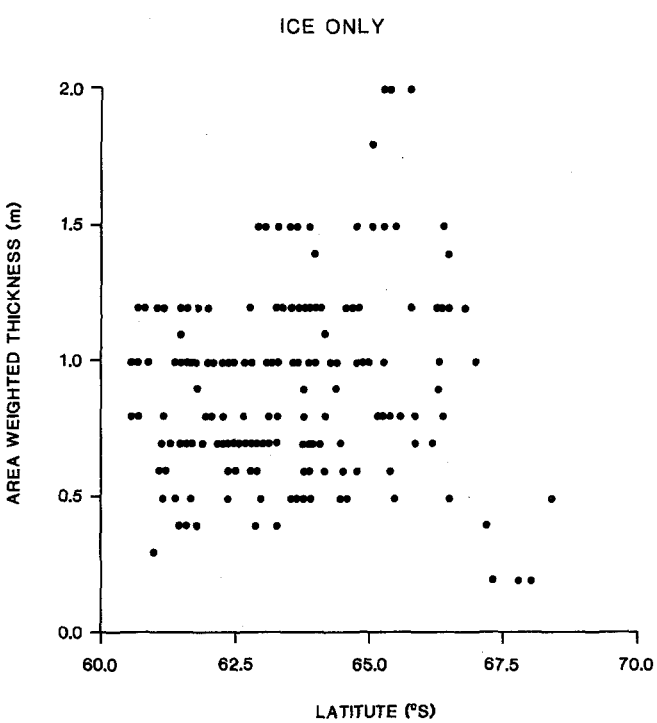


Fig 6 Latitudinal variation of the area weighted ice thickness (a) including open water and (b) for ice covered areas only

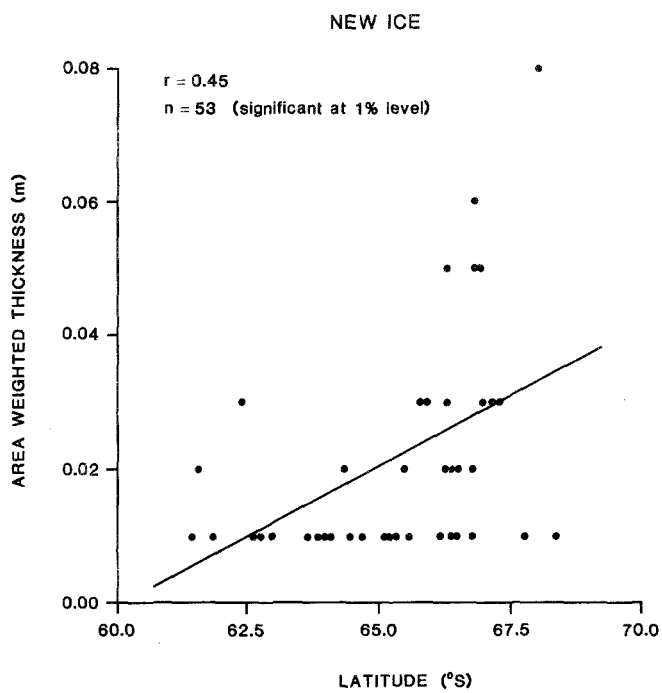


Fig 7 Latitudinal variation of the total area weighted thickness of new ice

concentration obtained from the 'Icebird' observations compared with the concentrations on the JIC charts. There is good agreement in both the general pattern and in the concentration levels, but the shipboard observations additionally showed that within the high concentration pack much of the ice was thin.

The lower parts of Fig 9 show the variation in total area weighted ice thickness both for all ice types and for first year ice only (> 0.3 m thick). The ice thickness decreases both with distance N and with time, while the shore polynya expands northward after early November since ice moving N is not replaced by new ice growth. Although young ice types were widespread, the small difference between the two lower parts of Fig 9 shows that they contribute only about 0.1 m thickness to the total ice mass. Very little melt would have occurred in the pack prior to the beginning of November and ice thickness during mid winter would not be much greater than that shown from the early observations.

The occurrence of extensive young ice in the pack was matched by the lack of significant ridging. Most of the floes were only moderately hummocked and only in the region around 64° S, 101° E, where ice drift was impeded by a larger (34×60 km), grounded iceberg that had recently calved from the Denman Glacier/Shackleton Ice Shelf system, was the ice under consistent pressure.

The Drift of East Antarctic Pack

The eastward surface ocean flow in the Antarctic Circumpolar Current (ACC) is at a maximum at the Polar Front (Hofman 1985) and decreases to the S. Ageostrophic Ekman transport results in a northward component of drift in the ACC, and Gordon and Taylor

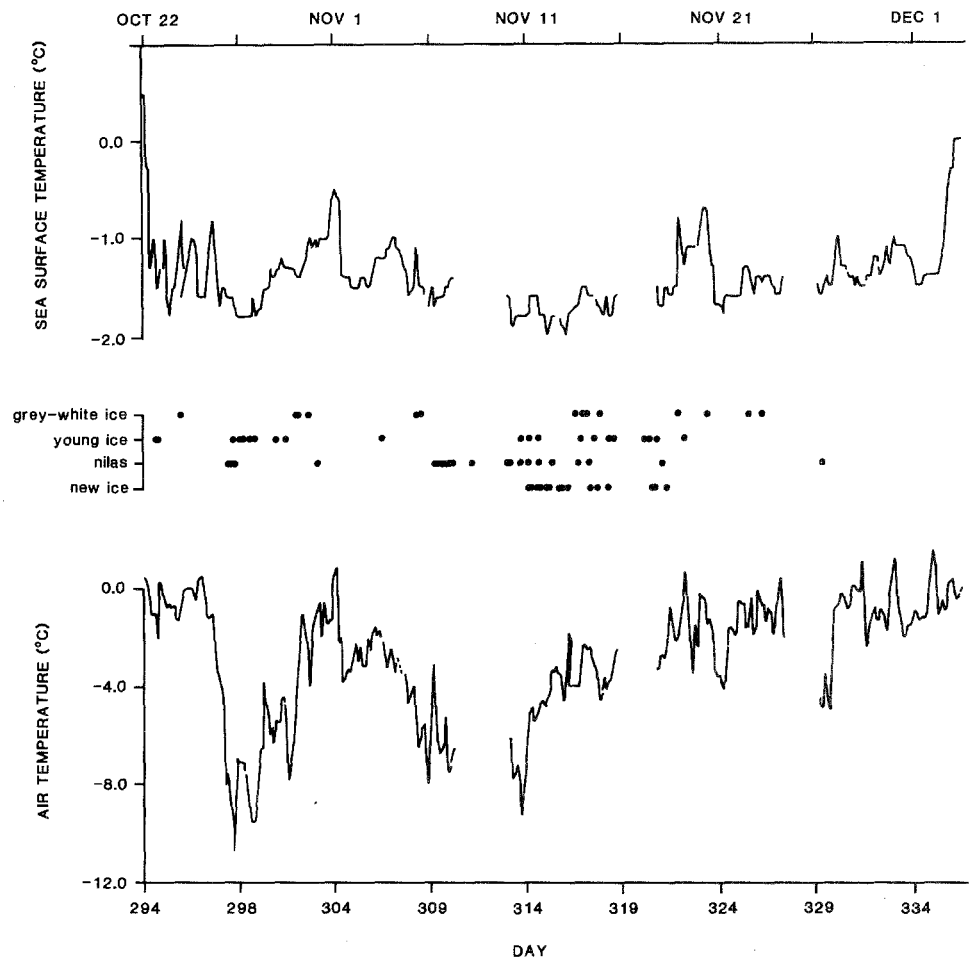


Fig 8
Air temperatures, sea surface temperatures and periods of young ice occurrence during the cruise of MV 'Icebird'

(1975) suggested that this divergence is a major cause of the large seasonal variation in Antarctic ice. South of the Antarctic Divergence, close to the Antarctic coast, the mean current is to the W (the East Wind Drift, EWD). From iceberg trajectories Tchernia and Jeannin (1984) identified a number of regions of northward transport linking the westerly and easterly surface flows. These include the Weddell and Ross Seas and regions around 90° E and 160° E, all areas where bottom topography influences surface flow in the low baroclinic Antarctic continental shelf water.

Over long time scales a major component of sea ice drift results from the mean ocean circulation, but on a day to day time scale the ice is primarily wind driven with additional forcing from water drag, Coriolis acceleration, and, in areas of ice convergence, the stress between floes.

Data buoy studies of Antarctic sea ice drift have previously been reported only for the Weddell Sea (Ackley 1979; Ackley and Holt 1983): recent data from drifting buoys studies in the East Antarctic region are presented here. In late February/early March 1985 three buoys (ICEX41 to 43) were deployed in Prydz Bay (70 to 80° E) and a further six buoys (ICEX44 to 49) were deployed in the same region in early March 1987. These buoys measured air temperature and pressure, sea surface temperature and ocean temperatures to 100 m depth, and transmitted data via the ARGOS positioning and data relay system on TIROS-N/NOAA satellites. The drift tracks of the buoys are shown in Fig 10. The tracks of the 1985 buoys are shown from the date of deployment until they ceased operation (ICEX41, 28 September 1985; ICEX42, 30 April 1986; ICEX43, 30 January 1986) while those for the 1987 buoys are shown from the date of deployment until 1 November 1987. One buoy (ICEX47) failed shortly after deployment and is not shown in Fig 10.

The buoys were deployed in open water and allowed to become entrapped within the advancing autumn ice edge. Temperature data from the buoys themselves and comparison of their position with the JIC ice charts shows that in 1985 all three buoys were caught in the pack for considerable periods but in 1987 only two buoys were in ice. The buoy trajectories when within ice are indicated on Fig 10.

Data from the buoys and their daily drift rate, which are discussed in more detail by Allison (in press), show a number of features. The buoys either moved westward

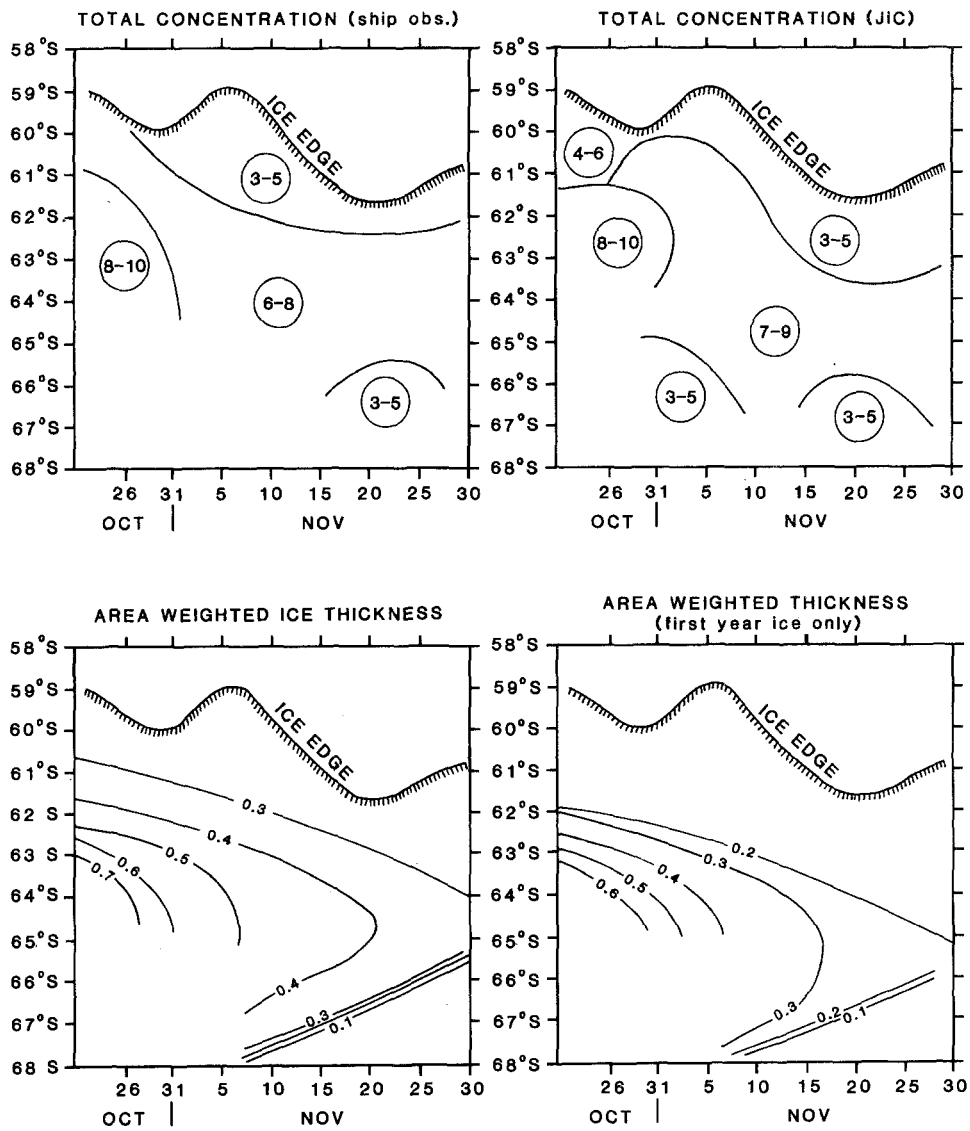


Fig 9
Changes of ice concentration and weighted ice thickness with latitude and time for the zone from 50° E to 120° E

around the coast in the EWD or eventually eastward in the ACC N of the ice edge. Those in the ACC first drifted northward at a rate that paralleled the autumn advance of the ice edge. This is illustrated in Fig 11. ICEX42 in particular remained at or near the ice edge for up to six months. Although the three northernmost 1987 buoys always stayed free of the pack, their northward drift also paralleled the ice edge advance. These trajectories suggest that ice divergence to the N plays an important role in determining the position and advance rate of the ice edge.

Data from the buoy temperature sensors and the day to day variablility of the drift provided an indication as to whether the buoys were in open water, low concentration ice in the marginal ice zone (MIZ), or 9–10/10 pack ice. When in ice the buoys had a high day to day variability in drift speed as the ice was primarily driven by the wind. In contrast, in open water, with the 100 m thermistor chain serving as a drogue, the buoys drifted with ocean currents and showed much smoother changes in drift rate. The air temperatures in open water were generally above 2° C, in the MIZ they were around −5° C, and in heavy ice they showed greater fluctuations and were often below −20° C. The classification of ice conditions according to the buoy drift rate and measured temperatures agreed well with the ice conditions shown on JIC charts at the bouy location.

In Tab 1 the average daily displacement and mean vector velocity are shown for the 1985 buoys for periods of different ice condition. The daily buoy displacements are generally high. Even when more than 500 km from the ice edge, in 9–10/10 concentration ice, buoy ICEX41 drifted an average of 14 km and as much as 55 km in a day. West of 20° E however, as the 1987 buoys approached the heavier ice in the Weddell Sea, the drift rates became much slower.

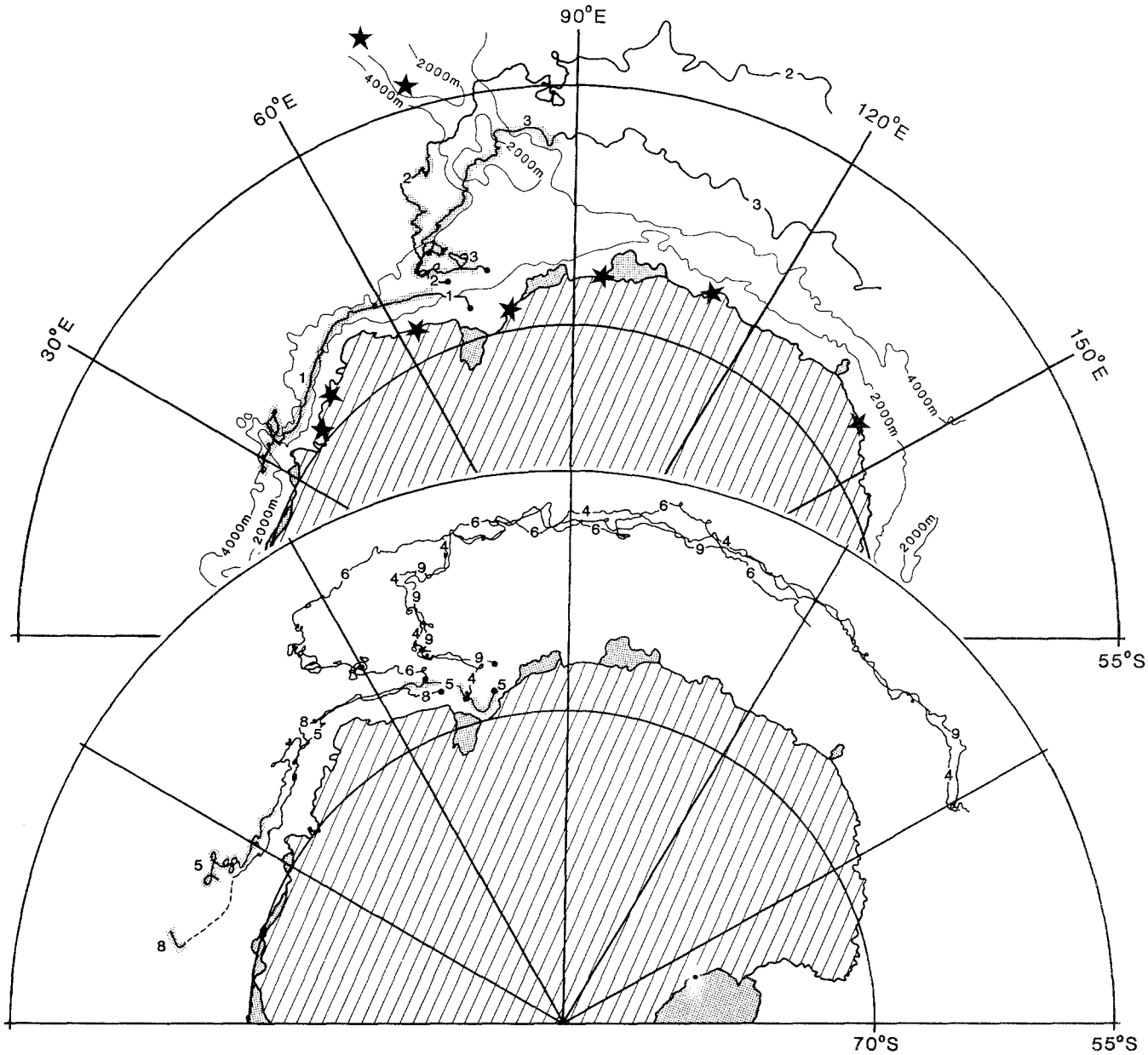


Fig 10 Drift tracks of data buoys deployed in Prydz Bay in February/March 1985 (top) and in March 1987 (bottom). Numbers on the tracks are the last digit of the buoy (e.g. 1 = ICEX41). Stippled tracks indicate those periods when the buoys were within the sea ice edge as shown on the JIC weekly ice charts. Also shown (top) are meteorological stations ( ) and the 2000 and 4000 m bathymetric contours.

The average vector speed in the EWD was about 0.07 ms^{-1}, similar to the average drift rate of icebergs (Tchernia and Jeannin 1984) and showing the importance of ocean currents in determining the long term ice drift. In the ACC the vector speed increased from 0.10 ms^{-1} at 60° S to 0.15 ms^{-1} at 55° S, similarly to the current speeds derived by Budd (1986) from FGGE buoys.

Allison (in press) investigated the relationship between average daily buoy drift and the 1000 hPa geostrophic wind interpolated from the Australian Bureau of Meteorology gridded Southern Hemisphere analyses (500×500 km). Because of lack of data in this region the wind speeds were too inaccurate to show any significant correlation with ice drift, but daily wind and buoy headings did show correlation when the buoys were in ice (Tab 1). In open water, with the buoys drifting with the mean ocean current, daily wind heading had no significant correlation with buoy heading. For buoys N of the Antarctic Divergence the buoy heading was to the left of the wind (Tab 1) as would be expected from Ekman divergence in a loose ice field. In the EWD however

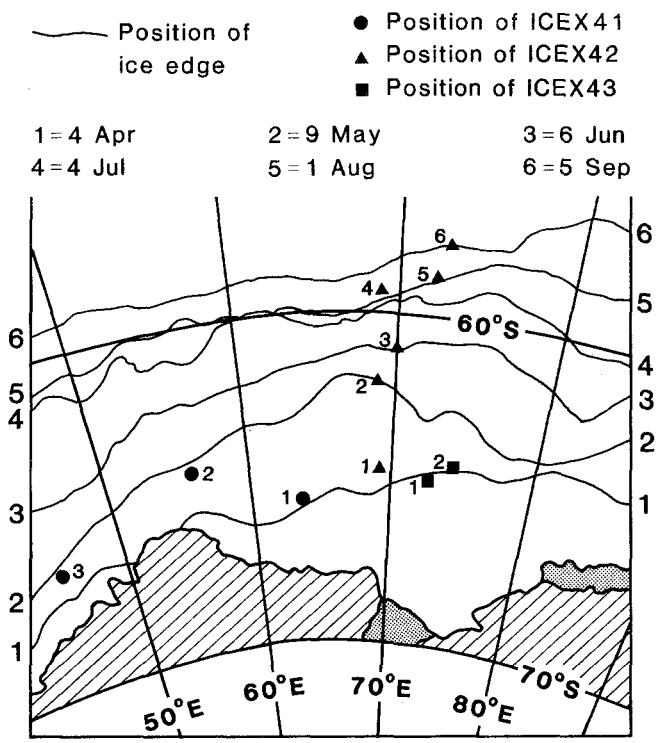


Fig 11 Position of drifting buoys relative to the JIC ice edge on indicated dates during the 1985 expansion of the ice cover

(ICEX41, May–September), the buoy drift was to the right of the wind because of interaction between ice floes in the convergent drift around the coast.

Concluding Remarks

The data presented here confirm the earlier observations of Jacka et al. (1987) showing that in late spring much of the sea ice in the Indian Ocean sector is relatively thin young ice. The spring ice thickness averaged over the total area was less than 0.4 m and even in winter the bulk of the ice is probably less than 1 m thick. The drift of Antarctic ice undoubtedly has a major influence on the mass budget of the pack and, particularly because of the generally divergent flow over most of the region, affects ice extent, ice concentration and ice type and hence the ocean-atmosphere energy balance.

The long term drift of the ice is determined by the mean ocean circulation but on a day to day time scale the ice is primarily wind driven. North of the Antarctic Divergence ship observations and buoy data suggest that the ice drift is generally divergent and that there is little stress between floes.

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Tab 1 Buoy drift speed and relationship between buoy and wind heading

	average distance made good (km)	mean vector buoy speed (km/day)	mean vector buoy heading (deg.)	mean vector wind heading (deg.)	coefficient of correlation (r) between daily buoy heading and wind heading
ICEX41					
MIZ (01/03/85-19/05/85)	16.3	11.2	277	308	0.81
9–10/10 ice (20/05/85-27/09/85)	14.3	5.8	269	241	0.83
ICEX42					
MIZ (03/03/85-07/10/85)	11.0	4.0	44	87	0.79
Open ocean (08/10/85-30/01/86)	28.3	12.4	70	84	0.22
ICEX43					
MIZ (03/03/85-30/04/85)	15.3	7.2	276	332	0.81
9–10/10 ice (01/05/85-12/09/85)	19.8	8.0	32	61	0.83
Open ocean (13/09/85-30/01/86)	13.9	9.8	96	92	0.37

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